
UNIMAS: A Unified View of Multi-Agent Systems for Embodied AI

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1 Workshop Description

Location The preferred location is **Atlanta, GA**. Some of our organizers have visa restrictions that favor a U.S.-based location.

Workshop Scope Multi-agent systems are central to real-world intelligent systems, including robotic teams, autonomous driving, manufacturing, and healthcare, where agents must coordinate under uncertainty and dynamic environments (Chen et al., 2019; Wu et al., 2025; Adil et al., 2025). In parallel, recent advances in AI have produced a new class of multi-agent systems driven by LLMs, VLMs, and world models, enabling complex reasoning, planning, and coordination in simulated environments (Pham et al., 2024; Li et al., 2023). However, these two paradigms remain fundamentally misaligned: **robotic control treats multi-agent systems as physically grounded, safety-critical dynamical systems, while modern AI treats them as representation-driven, reasoning-based agents**. Robotics emphasizes dynamics, constraints, and distributed control (Chen et al., 2019; Wu et al., 2025), whereas AI focuses on learned representations and flexible interaction, including MARL (Foerster et al., 2016; Sukhbaatar et al., 2016), emergent communication (Lazaridou et al., 2017, 2018; Mordatch & Abbeel, 2018), and LLM-based coordination (Li et al., 2023; Pham et al., 2024). At the same time, recent work reveals structural convergence across these paradigms, including latent message passing, symbolic communication, and parameter-level coordination (Foerster et al., 2016; Lazaridou et al., 2017; Terry et al., 2020; Dagan et al., 2021; Zhang et al., 2025; Tessera et al., 2025). **UNIMAS addresses this gap by reframing multi-agent systems as a unified problem of embodied intelligence across physical and digital domains, focusing on shared principles of representation, coordination, and distributed decision-making.**

Workshop Timeliness This gap is increasingly urgent as embodied AI pushes multi-agent systems from simulation to real-world deployment. LLM-based agents and world models enable scalable reasoning and coordination (Pham et al., 2024), while robotics and control provide robust mechanisms under physical constraints (Chen et al., 2019). However, no existing venue explicitly connects these paradigms at the level of system abstraction and design principles. UniMAS directly targets this missing layer.

Community Building UNIMAS builds a community that connects robotics, control, and embodied systems with LLM agents, world models, and multi-agent AI. It establishes a shared vocabulary, aligns assumptions, and surfaces core challenges in multi-agent intelligence. By unifying model-centric and physically grounded perspectives, UNIMAS enables meaningful cross-disciplinary collaboration and accelerates progress toward real-world multi-agent systems, making it a natural fit for NeurIPS.

Broader Outreach We have secured initial sponsorship (\$750 - \$1500 from Abaka AI) and are pursuing additional support from Cisco, Nvidia, the Commonwealth Cyber Initiative, and the Virginia Tech AI Center to fund student travel awards. Our team has prior experience managing such programs. UNIMAS will also support participation from **K-12 and undergraduate students from local and underrepresented communities** through targeted attendance awards, lowering barriers to entry and expanding the pipeline of future researchers in embodied multi-agent intelligence.

Past Workshops Recent workshops have explored adjacent themes, including *AI Agents* (COLM 2025), *Multi-Agent Systems in the Era of Foundation Models* (ICML 2025), *Multi-Agent Embodied Intelligent Systems* (CVPR 2025), *Embodied World Models for Decision Making* (NeurIPS 2025), and *REALM: Research on Agent Language Models* (ACL 2025). These workshops focus on specific directions, including LLM-based agents, embodied perception and control, or MARL and coordination, but **do not address the fundamental mismatch across communities in how multi-agent systems are defined, modeled, and evaluated**. UNIMAS differs by explicitly targeting this gap: it focuses on the *conceptual unification* of multi-agent systems across robotics/control and modern AI, grounded in a shared view of embodied intelligence.

2 Workshop Content

Topics of Interest We welcome submissions on topics spanning the full range of multi-agent systems for embodied AI. Topics of interest include, but are not limited to, the following:

- **Embodied Multi-Agent Systems** involving robotic teams, autonomous vehicles, and physical agents operating under real-world constraints
- **World Models for Agent Coordination** using simulators and predictive environments for planning and collaborative reasoning
- **Unifying Physical and Digital Agents** through shared abstractions connecting robotics and AI-driven reasoning systems
- **Benchmarks and Evaluation** for embodied multi-agent systems, including metrics, datasets, and experimental testbeds

Submission Timeline A tentative paper submission timeline is as follows:

- **Submission Deadline:** August 29, 2026, AoE
- **Acceptance Notification:** September 29, 2026, AoE
- **Workshop Date:** December 12 or 13, 2026

Submission Format UNIMAS solicits research paper submissions (9 pages main text + unlimited references) describing original, ongoing, or recently completed work at the intersection of multi-agent systems, embodied AI, and foundation-model-based agents. Submissions are **non-archival** and will be hosted on **OpenReview** to promote transparency and community engagement. Each paper will receive at least two **double-blind** reviews from program committee members with expertise spanning robotics/control and AI/LLM agents, ensuring balanced and interdisciplinary evaluation. Conflicts of interest will be managed according to standard NeurIPS/OpenReview policies, including automatic detection and manual verification by organizers. Submission timelines and decision deadlines will follow NeurIPS guidelines. Accepted papers will be presented as posters or oral talks at the workshop. Additional details are available on our **website**.

3 Workshop Format

UNIMAS will be a full-day program featuring **four keynote talks** spanning robotics/control and LLM-based agent systems. The program includes invited talks in the morning, followed by contributed oral presentations, a panel discussion with invited speakers and organizers on unifying multi-agent systems across paradigms, and a poster session for accepted papers. Coffee breaks and lunch will be provided. The workshop will conclude with a networking reception integrated with the poster session to facilitate discussion and collaboration. A tentative schedule is shown in Table 1 and our **website**.

Table 1: Tentative Workshop Schedule

Time	Activity	Description
08:30 – 08:45	Opening Remarks	Welcome and workshop overview
08:45 – 09:35	Keynote Talk 1	Invited keynote presentation
09:35 – 10:25	Keynote Talk 2	Invited keynote presentation
10:25 – 10:55	Coffee Break	Refreshments and networking
10:55 – 11:45	Keynote Talk 3	Invited keynote presentation
11:45 – 12:30	Keynote Talk 4	Invited keynote presentation
12:30 – 13:45	Lunch Break	Opportunity for attendees to connect
13:45 – 15:15	Contributed Talks (Oral)	Selected peer-reviewed papers
15:15 – 15:45	Coffee Break	Refreshments and networking
15:45 – 16:45	Panel Discussion	Discussion with invited speakers and organizers
16:45 – 17:00	Closing Remarks & Best Paper Award	Summary of key insights and award announcement
17:00 – 18:00	Networking Reception & Poster Session	Poster viewing and informal discussions

Invited Speakers Below is a list of **confirmed keynote speakers**:

- **Dawn Song** is a Professor at UC Berkeley whose research focuses on AI security, privacy, and trustworthy machine learning, with broad impact on real-world AI systems.
- **Sergey Levine** is an Associate Professor at UC Berkeley whose research focuses on deep reinforcement learning and robot learning to learn complex behaviors from data.
- **Jiajun Wu** is an Assistant Professor at Stanford whose research focuses on physical scene understanding, enabling machines to see, reason, and interact with the world.
- **Chi Wang** is a Senior Staff Research Scientist at Google DeepMind and the creator of AutoGen (AG2), with broad impact on agentic AI and real-world applications.

Number of Attendees We expect **50-100 participants** from academia and industry across robotics, control, embodied AI, and foundation-model-based agents. UNIMAS is uniquely positioned as the only workshop explicitly targeting the conceptual unification of multi-agent systems across physical and AI-driven paradigms, addressing a clear gap and attracting researchers seeking cross-disciplinary integration. Outreach will leverage academic channels, professional networks, and speaker visibility to ensure strong participation.

Diversity and Inclusion We are committed to promoting demographic diversity among organizers and speakers. Our team includes individuals from different career stages, institutions, and cultural backgrounds, with attention to gender balance and representation from both academia and industry. In our call for proposals, we will use inclusive language and explicitly encourage submissions from early-career researchers and underrepresented groups. We also plan to offer mentoring opportunities for junior participants and financial subsidies to help offset travel costs for those with limited resources.

Technical Requirements The workshop requires standard audiovisual support, including a projector, screen, laptop connections, and microphones (with at least one wireless microphone for audience interaction). For the poster session, we will need display boards or easels. Reliable Wi-Fi is important for engaging with online materials and collaboration. A timer or signaling system to help presenters stay on schedule would be appreciated. Given the interactive nature of the workshop, a flexible room layout that supports informal group discussions and networking will help create a dynamic and engaging environment for the participants.

4 Organizing Committee

We have composed a diverse organizing team with representation across gender, geographic regions (east to west coast), seniority levels (junior to senior researchers), academic and industry, and research domains (LLM, embodied AI, robotics, medical domains). All organizers **confirm** that they are **not listed in more than two NeurIPS 2026 workshop proposals**.

- **Xuan Wang (xuanw@vt.edu)** is an Assistant Professor in the Department of Computer Science at Virginia Tech. Her research focuses on developing scalable and reliable multi-agent foundation model systems for decision-making in complex multimodal real-world environments. She was a recipient of the NSF CAREER Award 2025, Nvidia Academic Grant 2025, Cisco Research Award 2025, NSF NAIRR Pilot Award 2024 - 2025, and NAACL Best Demo Paper Award 2021. Xuan has served as a Program Chair of the SouthNLP Symposium 2024 (>150 participants), from more than 20 universities across the USA. She has also served as a Program Chair for the Undergraduate and High School Symposium at IEEE-BigData 2024 and IEEE-ICDM 2025. She has also served as a Senior Area Chair or Area Chair in major AI conferences (e.g., ARR, ACL, EMNLP, NeurIPS).
- **Manling Li (manling.li@northwestern.edu)** is an assistant professor of Computer Science at Northwestern University, affiliated with Center for Robotics and Biosystems and Cognitive Science. She directs the Multimodal Learning with Language (MLL) Lab. She is also an Amazon Scholar working on conversational agents. She is a recipient of MIT Tech Review Innovators Under 35 in 2025, ACL Inaugural Best Dissertation Award Honorable Mention in 2025, DARPA Riser in 2022, and a EECS Rising Star in 2022, Microsoft Research PhD Fellowship in 2021, etc. Her work on multimodal reasoning was recognized as ACL'24 Outstanding Paper Award, ACL'20 Best Demo Paper Award, NAACL'21 Best Demo Paper Award, etc. She served as organizing committee of ACL'25 (virtual chairs), NAACL'25 (publication chairs), EMNLP'24 (demo chairs), etc.
- **Wenqi Shi (wenqi.shi@utsouthwestern.edu)** is an Assistant Professor in the Department of Health Data Science and Biostatistics at the University of Texas Southwestern Medical Center. Her research focuses on the intersection of AI and healthcare, advancing both foundational algorithms and agentic systems for precision and personalized medicine. Her recent work includes developing LLMs for translational medicine, advancing agentic AI for biomedical discovery, and promoting responsible AI practices to improve clinical research and care. She has received the NVIDIA Academic Grant Program Award (2025), the Texas Advanced Computing Center (TACC) Pilot Award (2025), the Best SIGBio Paper Award at ACM BCB (2023), and recognition as a Rising Star in EECS (2023). She has served as Finance Co-Chair of ACM BCB 2025, Publication Co-Chair of IEEE BHI 2025 (both > 300 participants), and as an Area Chair or Program Committee member in major AI and healthcare conferences (e.g., ACL, EMNLP, NAACL, NeurIPS, ICLR, ICML, COLM, BCB).
- **Yuchen Zhuang (yczhuang@google.com)** is a Research Scientist at Google DeepMind. His research focuses on language intelligence, with the goal of developing LLM-based agents that can show human-like reasoning and planning on challenging real-world tasks. His experience covers model pre-training, instruction fine-tuning, reinforcement learning from human feedback (RLHF), and evaluation. He has published more than 30 papers in top AI venues such as NeurIPS, ICLR, ICML, ACL, EMNLP, NAACL, UAI, and KDD. He has received the 2024 J.P. Morgan PhD Fellowship Award, the NeurIPS 2023 Scholar Award, and the Best SIGBio Paper Award at ACM BCB. He serves as an Area Chair and Program Committee member in major AI conferences (e.g., ARR, ACL, EMNLP, NAACL, NeurIPS, ICLR, ICML).
- **Ligeng Zhu (ligengz@nvidia.com)** is a Research Scientist at Nvidia Research. He leads research into building efficient, scalable, and agentic Large Language Models (LLMs) and Vision-Language Models (VLMs). Ligeng is a primary architect of the NVILA (Visual Language Models) family at NVIDIA. His work on VLMs redefined VLM pre-training with interleaved medias and texts, massive context windows for long-video understanding, and NVILA, a next-generation native VLM designed for high-resolution perception and efficient deployment in AI agent scenarios. He has published more than 40 papers in top AI venues such as NeurIPS, ICLR, ICML, ECCV, and CVPR. He has received the 2022 Qualcomm PhD Fellowship Award.
- **Charles Flemming (chflemm@cisco.com)** is a Senior Researcher at Cisco Research. Charles worked for ten years in industry before returning to graduate school. He completed his PhD in Computer Science at UCLA in 2013 and worked as a faculty member at Xi'an Jiaotong-Liverpool

University and the University of Mississippi before joining Cisco in 2022. Charles works broadly in security and privacy, with a special interest in machine learning applications in security and security for AI.

- **Zihan Wang (888423@gmail.com)** is the founding member and VP of Research at Abaka AI, where he leads global research and community collaborations on large-scale data infrastructure for frontier AI systems. He also heads the 2077AI Research Foundation, an open-source nonprofit focused on building community-driven benchmarks and evaluation frameworks for next-generation AGI models. Meanwhile, Zihan is deeply involved in community and thought leadership, organizing initiatives across top-tier computer science conferences (e.g., NeurIPS, CVPR, ICML, ICLR), workshops, and webinars.

5 Advising Committee

- **Heng Ji (hengji@illinois.edu)** is a Professor at Computer Science Department of University of Illinois Urbana-Champaign, and an Amazon Scholar. Her research interests focus on Natural Language Processing, especially on Multimedia Multilingual Information Extraction, Knowledge Base Population and Knowledge-driven Generation. She was selected as “Young Scientist” and a member of the Global Future Council on the Future of Computing by the World Economic Forum in 2016 and 2017. The awards she received include “AI’s 10 to Watch” Award by IEEE Intelligent Systems in 2013, NSF CAREER award in 2009, Google Research Award in 2009 and 2014, IBM Watson Faculty Award in 2012 and 2014 and Bosch Research Award in 2014-2018, Amazon AWS Award in 2021, ACL2020 Best Demo Paper Award, and NAACL2021 Best Demo Paper Award. She has coordinated the NIST TAC Knowledge Base Population task since 2010. She has served as the Program Committee Co-Chair of many conferences including NAACL-HLT2018. She is elected as the NAACL secretary 2020-2021.
- **Jiawei Han (hanj@illinois.edu)** is Michael Aiken Chair Professor, Department of Computer Science, University of Illinois at Urbana-Champaign. His research areas encompass data mining, text mining, data warehousing, and information network analysis, with over 1000 research publications. He is Fellow of ACM, Fellow of IEEE, and received numerous prominent awards, including ACM SIGKDD Innovation Award (2004) and IEEE Computer Society W. Wallace McDowell Award (2009). He delivered 50+ conference tutorials or keynote speeches (e.g., SIGKDD 2017-2021 tutorials and WSDM 2018 keynote).

6 Program Committee

Below is a list of our **confirmed** program committee members:

- Anil Vullikanti, Professor, Computer Science, University of Virginia
- Ashok Palaniappan, Associate Professor, SASTRA Deemed University, India
- Nikhil Muralidhar, Assistant Professor, Stevens Institute of Technology
- Jiaying Lu, Research Assistant Professor, Emory University
- Saurabh Kataria, Postdoctoral researcher, Emory University
- Huaming Chen, Senior Lecturer, University of Sydney
- Jun Hou, PhD Student, Virginia Tech
- Priya Pitre, PhD Student, Virginia Tech
- Xuxin Tang, PhD Student, Virginia Tech
- Junhao Liu, PhD Student, University of California, Irvine
- Junze Liu, PhD Student, University of California, Irvine
- Yuhui Hong, PhD Student, Indiana University
- Guangyao Chen, Postdoc, Cornell University
- Yingheng Wang, PhD Student, Cornell University
- Vardhan Dongre, PhD Student, University of Illinois Urbana-Champaign

- Satyapriya Krishna, PhD Student, Harvard University
- Xuemei Gu, Postdoctoral researcher, Max Planck Institute for the Science of Light
- Siyuan Yu, Master Student, University of Alberta
- Azmine Toushik Wasi, Graduate Student, Shahjalal University of Science and Technology
- Shan Xue, Senior Data Scientist & Computational Chemist, Amgen
- Amartya Hatua, Senior Data Scientist, Fidelity Investments
- Yichun He, Eric and Wendy Schmidt Fellow, Broad Institute of MIT and Harvard
- Ashwin Samudre, Scientific Engineer, Allen Institute

7 Conflict of Interest

We are committed to upholding a transparent and rigorous conflict of interest (COI) policy in full compliance with NeurIPS guidelines. To this end, we will adhere to the following rules:

- Organizers and committee members will disclose any other potential conflicts, including collaborations, advisor-advisee relationships, or financial interests, and abstain from involvement in related submissions.
- Organizers and program committee members will recuse themselves from reviewing or making decisions on submissions authored by individuals from their own institutions or with whom they have close personal or professional relationships.
- Organizers will not give contributed talks at the workshop to avoid conflicts in dual roles; they may only provide brief introductions or moderate sessions.
- We will strictly follow the NeurIPS COI policies, ensuring that all conflicts are handled proactively to maintain fairness and integrity throughout the workshop planning and review process.
- We will maintain a COI management record and ensure that no organizer or reviewer assesses submissions where a conflict exists.

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